

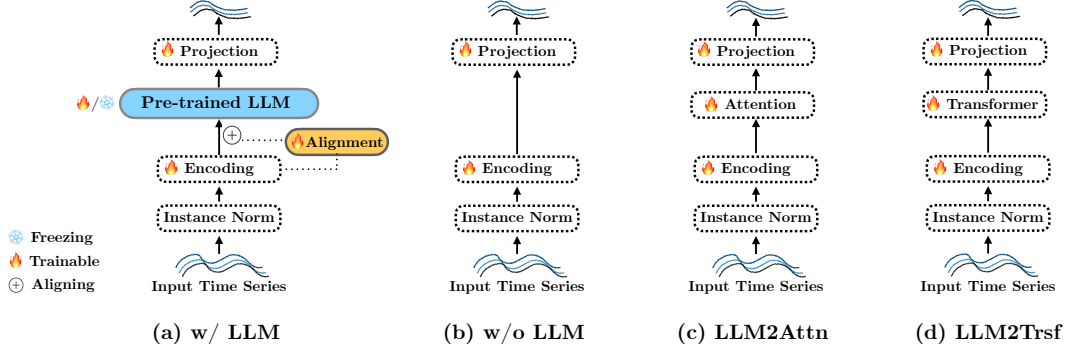


Figure 1: Overview of all LLM ablation methods. Figure (a) represents time series forecasting using an LLM as the base model. In some works, the LLM components are frozen [14, 13], while in others, they undergo fine-tuning [49, 21, 4]. Figure (b) shows the model with the LLM components removed, retaining only the remaining structure. Figure (c) replaces the LLM components with a single-layer self-attention mechanism. Figure (d) replaces the LLM components with a simple Transformer.

3.2 Proposed Ablations

To isolate the influence of the LLM in an LLM-based forecaster, we propose three ablations: removing the LLM component or replacing it with a simple block. Specifically, for each of the three methods we make the following three modifications:

- **w/o LLM** (Figure 1 (b)). We remove the language model entirely, instead passing the input tokens directly to the reference method’s final layer.
- **LLM2Attn** (Figure 1 (c)). We replace the language model with a single randomly-initialized multi-head attention layer.
- **LLM2Trsf** (Figure 1 (d)). We replace the language model with a single randomly-initialized transformer block.

In the above ablations, we keep left parts of the forecasters unchanged (trainable). For example, as shown in Figure 1 (a), after removing the LLM, the input encodings are passed directly to the output projection. Alternatively, as shown in Figure 1 (b) or (c), after replacing the LLM with attention or a transformer, they are trained along with the remaining structure of the original method.

3.3 Datasets and Evaluation Metrics

Benchmark Datasets. We evaluate on the following real-world datasets: (1) **ETT** [20]: encompasses seven factors related to electricity transformers across four subsets: ETTh1 and ETTh2, which have hourly recordings, and ETTm1 and ETTm2, which have recordings every 15 minutes; (2) **Illness** [39]: includes the weekly recorded influenza illness among patients from the Centers for Disease Control, which describes the ratio of patients seen with influenza-like illness to the total number of patients; (3) **Weather** [39]: local climate data from 1,600 U.S. locations, between 2010 and 2013, and each data point consists of 11 climate features; (4) **Traffic** [39]: is an hourly dataset from California transportation department, and consists of road occupancy rates measured on San Francisco Bay area freeways; (5) **Electricity** [34]: contains the hourly electricity consumption of 321 customers from 2012 to 2014. The train-val-test split for ETT datasets is 60%-20%-20%, and for Illness, Weather, and Electricity datasets is 70%-10%-20% respectively. The statistics for all datasets is given in Table 1. We highlight that these datasets, with the same splits and size, have been extensively used to evaluate time-series forecasting ability of LLM-based and other neural models for time-series data [47, 49, 4, 14, 5, 45, 39, 48]. (6) **Exchange Rate** [17]: collected between 1990 and 2016, it contains daily exchange rates for the currencies of eight countries (Australia, British, Canada, Switzerland, China, Japan, New Zealand and Singapore). (7) **Covid Deaths** [12]: contains daily statistics of COVID-19 deaths in 266 countries and states between January and August 2020. (8) **Taxi (30 min)** [1]: contains taxi rides from 1,214 locations in New York City between January 2015 and January 2016. The data is collected every 30 minutes, with an average of 1,478 samples. (9) **NN5 (Daily)** [12]: contains daily cash withdrawal data from 111 ATMs in the UK, with each ATM having

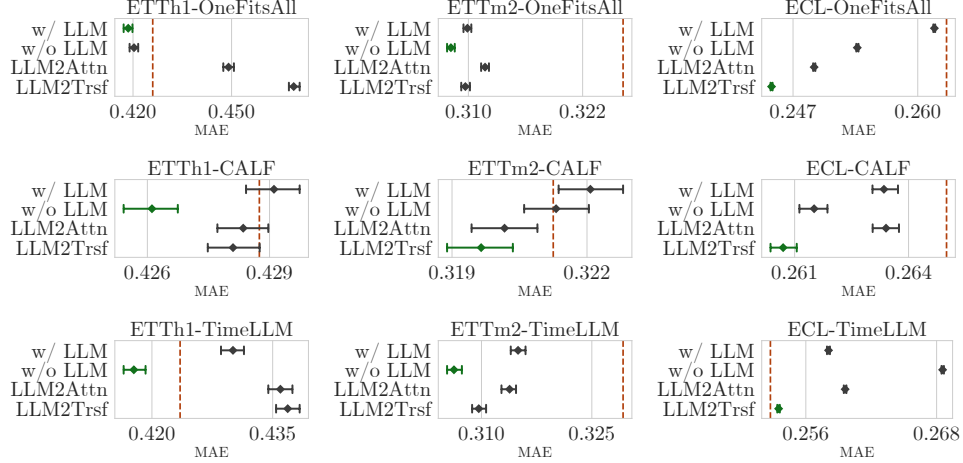


Figure 2: In the above examples, only OneFitsAll “w/ LLM” performs better than the ablation methods on ETTh1, but there is substantial overlap in bootstrapped confidence intervals. The figures show the comparison of OneFitsAll, CALF, and Time-LLM using LLMs and ablations (*i.e.*, **w/o LLM**, **LLM2Attn**, and **LLM2Trsf**) on ETTh1, ETTm2, and Electricity, and the vertical dashed lines represent the results from the original work. Others Figures for MSE and other datasets are available in Figure 5 and Figure 6 in the Appendix.

791 data points. (10) **FRED-MD** [12]: contains 107 monthly macroeconomic indices released by the Federal Reserve Bank since 01/01/1959. It was extracted from the FRED-MD database.

Evaluation Metrics and Setup. We report the results in terms of mean absolute error (MAE) and mean squared error (MSE) between predicted and true values of the time-series. Mathematically, given a test-set with $\mathcal{D}$ elements, $\text{MAE} = \frac{1}{|\mathcal{D}|} \sum_{t_i \in \mathcal{D}} |c_i - \hat{c}_i|$ and $\text{MSE} = \frac{1}{|\mathcal{D}|} \sum_{t_i \in \mathcal{D}} (c_i - \hat{c}_i)^2$, where c_i and $\hat{c}_i$ denote the true value and predicted value at the i -th index of the time-series respectively.

4 Results

In this section, we provide the details of our comprehensive evaluation of all baseline LLM models for time-series forecasting. Specifically, we ask the following research questions. **(RQ1)** Do pretrained language models contribute to forecasting performance? **(RQ2)** Are LLM-based methods worth the computational cost? **(RQ3)** Does language model pretraining help performance on forecasting tasks? **(RQ4)** Do LLMs represent sequential dependencies in time series? **(RQ5)** Do LLMs help with few-shot learning? **(RQ6)** Where does the performance come from?

4.1 Do pretrained language models contribute to forecasting performance? (RQ1)

Our results show that pretrained LLMs **are not useful for time series forecasting tasks yet**. Overall, as shown in Table 3, across 13 datasets and two metrics, ablations outperform Time-LLM methods in 26/26 cases, CALF in 22/26 cases, and OneFitsAll in 19/26 cases. We averaged results over different predicting lengths, as in [49, 14, 21]. Across all prediction lengths (thirteen datasets and four prediction lengths) ablations outperformed Time-LLM, CALF, and OneFitsAll in 35/40, 31/40, and 29/40 cases as measured by MAE, respectively. To ensure a fair comparison, we also report results from each method’s original paper alongside our replication. For specific results refer to Appendix E.1. To better evaluate the effectiveness of LLMs and ablation methods, we include 95% bootstrapped confidence intervals for each task. In tasks where LLMs performed better, such as OneFitsAll with ETTh1, shown in Figure 2, there is still substantial overlap in the confidence intervals with the ablation method “w/o LLM” in MAE. Other datasets results and MSE metrics are shown in Figure 5 and Figure 6 in the Appendix. To summarize, our results on the evaluation above, it is hard to conclude that LLMs are effective in time series forecasting.

4.2 Are LLM-based methods worth the computational cost? (RQ2)

In the previous section, we showed that LLMs do not meaningfully improve performance on time series forecasting tasks. Here, we evaluate the computational intensity of these methods with their

	Model → Dataset ↓	Time-LLM		w/o LLM		LLM2Attn		LLM2Trsf		From Original Paper	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Time-LLM [14]	ETTh1	0.432	0.417	0.419	0.405	0.437	0.422	0.439	0.429	0.423	0.408
	ETTh2	0.396	0.360	0.383	0.345	0.389	0.353	0.394	0.359	0.383	0.334
	ETTM1	0.377	0.356	0.371	0.350	0.376	0.356	0.377	0.359	0.371	0.329
	ETTM2	0.315	0.260	0.307	0.252	0.314	0.259	0.310	0.253	0.329	0.250
	Illness	0.894	2.017	0.924	1.956	0.849	1.789	0.837	1.795	0.801	1.435
	Weather	0.270	0.243	0.272	0.243	0.254	0.224	0.254	0.226	0.257	0.225
	Traffic	0.281	0.421	0.295	0.428	0.276	0.416	0.275	0.416	0.263	0.387
	Electricity	0.259	0.164	0.269	0.171	0.260	0.167	0.254	0.161	0.252	0.158
	Exchange Rate	0.448	0.422	0.413	0.384	0.432	0.403	0.442	0.422	-	-
	Covid Deaths	0.089	0.189	0.080	0.198	0.058	0.086	0.054	0.079	-	-
	Taxi (30 Min)	0.277	0.163	0.286	0.176	0.269	0.157	0.255	0.141	-	-
	NN5 (Daily)	0.432	0.402	0.425	0.379	0.411	0.364	0.401	0.347	-	-
	FRED-MD	0.0004	5e-7	0.0002	3e-7	0.0046	2.53e-5	0.0008	2.6e-6	-	-
	# Wins	0		12		2		12		-	
	#Parameters	6651.82M		0.55M		0.55M		0.66M		-	
	Model → Dataset ↓	CALF		w/o LLM		LLM2Attn		LLM2Trsf		From Original Paper	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
CALF [21]	ETTh1	0.431	0.431	0.428	0.436	0.430	0.428	0.430	0.430	0.428	0.432
	ETTh2	0.383	0.351	0.383	0.352	0.382	0.349	0.383	0.350	0.382	0.349
	ETTM1	0.391	0.396	0.390	0.397	0.390	0.396	0.390	0.394	0.390	0.395
	ETTM2	0.323	0.283	0.322	0.282	0.321	0.281	0.320	0.281	0.321	0.281
	Illness	0.869	1.699	0.861	1.639	0.892	1.748	0.860	1.630	-	-
	Weather	0.273	0.251	0.277	0.257	0.279	0.258	0.277	0.255	0.274	0.250
	Traffic	0.284	0.443	0.278	0.439	0.275	0.430	0.271	0.426	0.281	0.439
	Electricity	0.266	0.175	0.262	0.174	0.264	0.175	0.261	0.172	0.265	0.175
	Exchange Rate	0.417	0.388	0.409	0.367	0.417	0.389	0.409	0.367	-	-
	Covid Deaths	0.084	0.163	0.066	0.115	0.131	0.431	0.066	0.106	-	-
	Taxi (30 Min)	0.258	0.142	0.264	0.147	0.267	0.150	0.267	0.150	-	-
	NN5 (Daily)	0.403	0.362	0.386	0.336	0.463	0.445	0.415	0.381	-	-
	FRED-MD	0.0012	2.9e-6	0.0011	2.7e-6	0.0015	4.9e-6	0.0017	4.5e-6	-	-
	# Wins	4		7		4		11		-	
	#Parameters	180.25M		8.17M		10.5M		13.68M		-	
	Model → Dataset ↓	OneFitsAll		w/o LLM		LLM2Attn		LLM2Trsf		From Original Paper	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
OneFitsAll [49]	ETTh1	0.420	0.417	0.422	0.417	0.452	0.465	0.474	0.525	0.426	0.427
	ETTh2	0.388	0.353	0.389	0.356	0.402	0.375	0.397	0.367	0.394	0.354
	ETTM1	0.377	0.362	0.374	0.358	0.379	0.369	0.379	0.369	0.383	0.351
	ETTM2	0.310	0.253	0.309	0.255	0.312	0.256	0.310	0.254	0.326	0.266
	Illness	0.852	1.871	0.924	1.960	0.829	1.763	0.850	1.830	0.903	1.925
	Weather	0.254	0.226	0.272	0.245	0.256	0.226	0.256	0.228	0.270	0.236
	Traffic	0.273	0.420	0.273	0.439	0.266	0.415	0.256	0.409	0.294	0.414
	Electricity	0.262	0.169	0.254	0.165	0.250	0.162	0.245	0.157	0.263	0.166
	Exchange Rate	0.378	0.357	0.361	0.323	0.376	0.350	0.393	0.387	-	-
	Covid Deaths	0.057	0.075	0.050	0.073	0.058	0.103	0.078	0.162	-	-
	Taxi (30 Min)	0.252	0.138	0.259	0.143	0.259	0.145	0.257	0.140	-	-
	NN5 (Daily)	0.438	0.438	0.422	0.385	0.423	0.390	0.420	0.386	-	-
	FRED-MD	0.0006	1.2e-6	0.0002	4e-7	0.0006	1.5e-6	0.0012	2.4e-6	-	-
	# Wins	7		11		3		5		-	
	#Parameters	91.36M		9.38M		10.71M		13.54M		-	

Table 3: Forecasting performance of all models – Time-LLM, CALF, and OneFitsAll and results from our ablations. All results are averaged across different prediction lengths, though full results are available in Appendix E.1. Results in **Red** denote the best-performing model. # Wins refers to the number of times the method performed best, and # Params is the number of model parameters. “-” means the dataset is not included in the original paper.

nominal performance in mind. The language models in our reference methods use hundreds of millions and sometimes billions of parameters to perform time series forecasting. Even when the parameters of the language models are frozen they still contribute to substantial overhead during training and inference. For instance, Time-LLM has 6642 M parameters and takes 3003 minutes to train on the Weather dataset whereas ablation methods have only 0.245 M parameters and take 2.17 minutes on average. Information about training other methods on ETTh1 and Weather datasets are shown in Table 4. In the case of inference time, we divide by the maximum batch size to give an estimate of inference time per example. Time-LLM, OneFitsAll, and CALF take, on average, **28.2**,